

Characters of finite groups: complete proofs and exactly certified character tables of thirteen small groups

Hack-Nation Team
Verifier-gated discovery lab

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Abstract

We give a self-contained account of the complex representation theory of finite groups, from the definition of a representation to the column orthogonality relations, with every proof written out and only standard linear algebra assumed. The core results are Maschke's theorem, Schur's lemma, the identity $\dim \operatorname{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$, the formula $\sum_i (\dim V_i)^2 = |G|$ and the equality of the number of irreducible representations with the number of conjugacy classes. Every general statement is in addition tested on concrete groups by a verifier that works in exact arithmetic (rational numbers and cyclotomic fields, no floating point): we certify complete character tables for $C_1, \dots, C_8$, S_3 , D_4 , Q_8 , A_4 and S_4 , check $\dim \operatorname{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$ independently by Gaussian elimination on 52 pairs of representations, and certify that D_4 and Q_8 share a character table without being isomorphic. The verifier must first pass a self-test with 7 true and 8 false statements. The results are classical; the contribution is a short, complete and machine-reproducible exposition. The general proofs are not formally verified.

1 Introduction

Representation theory studies a group through its actions by linear maps. For finite groups over $\mathbb{C}$ the theory is complete in a strong sense: every representation splits into irreducible pieces, and a single function, the character, determines a representation up to isomorphism. The standard references are Serre [3], Fulton and Harris [1] and James and Liebeck [2]. This note collects the minimum needed to compute and use character tables, proves every step, and couples each general statement to computations that a reader can re-run in about one second with the Python standard library.

Each theorem-like environment carries a claim identifier in the margin. Identifiers starting with **E-** denote statements certified by exact computation (level `computed-rigorous`); the others denote statements whose proof is given in the text (level `proved-text`), each linked to the certificates that test it on examples (Appendix B).

Summary of results.

- Theorem 3 (Maschke) and Theorem 5 (Schur) give complete reducibility and the structure of G -maps between irreducible representations.
- Theorem 9 shows $\dim \operatorname{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$; Corollary 11 derives orthonormality of irreducible characters, multiplicities, and the criterion $\langle \chi, \chi \rangle = 1$ for irreducibility.

- Proposition 12 gives $\sum_i (\dim V_i)^2 = |G|$; Theorem 13 shows that the irreducible characters form an orthonormal basis of class functions; Theorem 15 gives column orthogonality.
- Examples 17–22 contain certified character tables of 13 groups; Proposition 23 shows that the character table does not determine the group.

2 Setting and notation

G is a finite group with identity e . Vector spaces are finite-dimensional over $\mathbb{C}$. We use without proof: (LA1) every linear endomorphism of a nonzero complex vector space has an eigenvalue; (LA2) a matrix whose minimal polynomial has no repeated roots is diagonalizable; (LA3) $\text{tr}(AB) = \text{tr}(BA)$; (GT) Lagrange's theorem on cosets.

Definition 1. A *representation* of G is a vector space V with a homomorphism $\rho: G \rightarrow GL(V)$; we write $gv = \rho(g)v$. A subspace $W \subseteq V$ is *G -invariant* if $gW \subseteq W$ for all g . V is *irreducible* if $V \neq 0$ and $0, V$ are its only invariant subspaces. A linear map $f: V \rightarrow W$ is a *G -map* if $f(gv) = gf(v)$; these form $\text{Hom}_G(V, W)$, and a bijective G -map is an isomorphism. $V^G = \{v : gv = v \forall g\}$.

Definition 2. The *character* of V is $\chi_V(g) = \text{tr } \rho(g)$. A function $f: G \rightarrow \mathbb{C}$ is a *class function* if $f(hgh^{-1}) = f(g)$. For $f_1, f_2: G \rightarrow \mathbb{C}$ put

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)},$$

a Hermitian inner product on class functions. The *regular representation* $\mathbb{C}[G]$ has basis $(e_x)_{x \in G}$ and $ge_x = e_{gx}$. The space $\text{Hom}(V, W)$ of all linear maps is a representation via

$$g \cdot f = \rho_W(g) f \rho_V(g)^{-1};$$

its fixed points are exactly $\text{Hom}_G(V, W)$. $C_G(g) = \{h : hg = gh\}$ is the centralizer.

3 Results

3.1 Linear algebra and characters

Lemma 1. (a) If $P^2 = P$ then $\text{tr } P = \dim \text{im } P$. (b) For $A \in \text{End}(W)$, $B \in \text{End}(V)$ the map $\Phi(f) = AfB$ on $\text{Hom}(V, W)$ has $\text{tr } \Phi = \text{tr } A \text{tr } B$. (c) If $A^m = I$ then A is diagonalizable with eigenvalues of modulus one and $\text{tr}(A^{-1}) = \overline{\text{tr } A}$. L-1

Proof. (a) $v = Pv + (v - Pv)$ with $P(v - Pv) = 0$, and $\text{im } P \cap \ker P = 0$ since $v = Pu$ gives $Pv = v$. In an adapted basis $P = \text{diag}(1, \dots, 1, 0, \dots, 0)$ with $\dim \text{im } P$ ones. (b) With matrix units, $AE_{ij}B = \sum_{k,l} A_{ki}B_{jl}E_{kl}$; the coefficient of E_{ij} is $A_{ii}B_{jj}$; sum over i, j . (c) The minimal polynomial divides $x^m - 1$, which has m distinct roots, so A is diagonalizable by (LA2) with eigenvalues λ_k , $\lambda_k^m = 1$, $|\lambda_k| = 1$, hence $\text{tr } A^{-1} = \sum \lambda_k^{-1} = \sum \overline{\lambda_k}$. □

Proposition 2. For representations V, W : (i) $\chi_V(e) = \dim V$; (ii) χ_V is a class function; (iii) $\chi_V(g^{-1}) = \overline{\chi_V(g)}$; (iv) $\chi_{V \oplus W} = \chi_V + \chi_W$; (v) *isomorphic representations have equal characters*; (vi) the character of $\text{Hom}(V, W)$ is $g \mapsto \chi_W(g) \overline{\chi_V(g)}$. P-1

Proof. (i) $\rho(e) = I$. (ii) and (v) by $\text{tr}(TAT^{-1}) = \text{tr } A$. (iii) $\rho(g)^m = I$ for m the order of g ; Lemma 1(c). (iv) ρ is block diagonal in an adapted basis. (vi) Lemma 1(b) with $A = \rho_W(g)$, $B = \rho_V(g^{-1})$, then (iii). □

3.2 Complete reducibility and Schur's lemma

Theorem 3 (Maschke). *Every G -invariant subspace $W \subseteq V$ has a G -invariant complement. Hence every representation is a direct sum of irreducible representations.* T-1

Proof. Let π be any linear projection of V onto W and $P = \frac{1}{|G|} \sum_g \rho(g) \pi \rho(g)^{-1}$. Since W is invariant, each summand maps V into W and fixes W pointwise; so $\text{im } P = W$ and $P^2 = P$. Reindexing $g \mapsto hg$ gives $\rho(h)P\rho(h)^{-1} = P$, so P is a G -map and $\ker P$ is invariant; $V = W \oplus \ker P$ as in Lemma 1(a). The second statement follows by induction on $\dim V$. □

Example 4. Let S_3 permute the coordinates of $\mathbb{C}^3$, $W = \mathbb{C}(1, 1, 1)$, and let E be the projection onto W along $\text{span}(e_1, e_2)$; E is not a G -map. The averaged projection of Theorem 3 is $P = \frac{1}{3}J$ with J the all-ones matrix: an invariant projection with image W and kernel $\{x_1 + x_2 + x_3 = 0\}$. *Certificate.* P is recomputed from E in exact rational arithmetic and $P^2 = P$, $P\rho(g) = \rho(g)P$ for all six g , $P(1, 1, 1)^T = (1, 1, 1)^T$ and $P(1, -1, 0)^T = P(0, 1, -1)^T = 0$ are checked. E-M

Theorem 5 (Schur). *Let V, W be irreducible and $f \in \text{Hom}_G(V, W)$. (a) $f = 0$ or f is an isomorphism. (b) If $V = W$ then $f = \lambda \text{id}$. Hence $\dim \text{Hom}_G(V, W)$ is 1 if $V \cong W$ and 0 otherwise.* T-2

Proof. (a) $\ker f$ and $\text{im } f$ are invariant; by irreducibility $f \neq 0$ forces $\ker f = 0$ and $\text{im } f = W$. (b) f has an eigenvalue λ by (LA1); $f - \lambda \text{id}$ is a G -map with nonzero kernel, hence 0 by (a). If $T: V \rightarrow W$ is an isomorphism, every $f \in \text{Hom}_G(V, W)$ satisfies $T^{-1}f = \lambda \text{id}$, so $\text{Hom}_G(V, W) = \mathbb{C}T$. □

Corollary 6. *Every irreducible representation of an abelian group has dimension 1.* C-1

Proof. Each $\rho(g)$ commutes with all $\rho(h)$, so it is a G -map and a scalar by Theorem 5(b). Then every subspace is invariant, and irreducibility forces dimension 1. □

3.3 Orthogonality

Lemma 7. $\dim V^G = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$. L-2

Proof. $P = \frac{1}{|G|} \sum_g \rho(g)$ satisfies $\rho(h)P = P$, so $\text{im } P \subseteq V^G$, and $Pv = v$ on V^G . Thus $P^2 = P$, $\text{im } P = V^G$, and $\dim V^G = \text{tr } P$ by Lemma 1(a). □

Example 8. For a permutation representation $\dim V^G$ is the number of orbits. *Certificate.* For S_3 , D_4 , A_4 , S_4 on their natural point sets the average of χ is 1, and for S_3 acting on $\{0, \dots, 4\}$ (fixing 3 and 4) it is 3; in each case it equals the number of orbits counted combinatorially. E-L2-S3,
E-L2-D4,
E-L2-A4,
E-L2-S4,
E-L2-S3x

Theorem 9. *For representations V, W of G : $\dim \text{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$.* T-3

Proof. Apply Lemma 7 to $\text{Hom}(V, W)$, whose fixed space is $\text{Hom}_G(V, W)$ and whose character is $\chi_W \overline{\chi_V}$ by Proposition 2(vi). □

Example 10. *Certificate.* For all ordered pairs among the irreducible representations of S_3 (resp. S_4) from Examples 18, 22 together with the permutation representation (16 resp. 36 pairs), $\dim \text{Hom}_G(V, W)$ is computed by Gaussian elimination over $\mathbb{Q}$ from $\rho_W(s)X = X\rho_V(s)$ for the generators s , without characters, and equals $\langle \chi_W, \chi_V \rangle$. In particular $\dim \text{Hom}_G(\mathbb{C}^n, \mathbb{C}^n) = 2$ for the permutation representation of S_n , $n = 3, 4$. E-T3-S3,
E-T3-S4

Corollary 11. Let $V_1, \dots, V_r$ be pairwise non-isomorphic irreducible representations with characters χ_i . (a) $\langle \chi_i, \chi_j \rangle = \delta_{ij}$. (b) If $V \cong \bigoplus_i V_i^{m_i}$ then $m_i = \langle \chi_V, \chi_i \rangle$; in particular the m_i do not depend on the decomposition. (c) Representations with equal characters are isomorphic. (d) $\langle \chi_V, \chi_V \rangle = \sum_i m_i^2$, and V is irreducible iff $\langle \chi_V, \chi_V \rangle = 1$. (e) Characters $\chi \neq \chi'$ with $\langle \chi, \chi \rangle = \langle \chi', \chi' \rangle = 1$ and $\langle \chi, \chi' \rangle = 0$ belong to non-isomorphic irreducible representations. c-2

Proof. (a) Theorems 9 and 5; since $\langle \chi_W, \chi_V \rangle$ is a nonnegative integer, the order of the arguments is irrelevant. (b) $\chi_V = \sum_j m_j \chi_j$ by Proposition 2; pair with χ_i . (c) By Theorem 3 and (b). (d) Expand with (a); conversely $\sum m_i^2 = 1$ with $m_i \in \mathbb{Z}_{\geq 0}$ leaves one summand. (e) Irreducible by (d); if isomorphic, the characters would be equal by Proposition 2(v), and then $\langle \chi, \chi' \rangle = 1$. $\square$

3.4 Counting irreducible representations

Proposition 12. (a) $\chi_{\text{reg}}(e) = |G|$ and $\chi_{\text{reg}}(g) = 0$ for $g \neq e$. (b) Every irreducible V occurs in $\mathbb{C}[G]$ with multiplicity $\dim V$. (c) There are finitely many irreducible representations $V_1, \dots, V_h$ up to isomorphism, and $\sum_i (\dim V_i)^2 = |G|$. (d) Pairwise non-isomorphic irreducible $V_1, \dots, V_r$ with $\sum_i (\dim V_i)^2 = |G|$ form a complete list. P-2

Proof. (a) $\rho(g)$ permutes the basis and fixes e_x iff $g = e$. (b) $\langle \chi_{\text{reg}}, \chi_V \rangle = \frac{1}{|G|} |G| \dim V$; Corollary 11(b). (c) By (b) each irreducible is a summand of $\mathbb{C}[G]$; compare dimensions. (d) A missing irreducible would add at least 1 to the sum in (c). $\square$

Theorem 13. The irreducible characters form an orthonormal basis of the space of class functions. In particular the number of irreducible representations equals the number of conjugacy classes. T-4

Proof. Let f be a class function with $\langle f, \chi_i \rangle = 0$ for all i . For a representation V set $T = \sum_g f(g) \rho(g)$. Then $\rho(h) T \rho(h)^{-1} = \sum_g f(h^{-1} g h) \rho(g) = T$, so T is a G -map. On $V = V_i$, Theorem 5 gives $T = \lambda \text{id}$ with $\lambda \dim V_i = \text{tr } T = |G| \langle f, \chi_i \rangle = 0$. Hence $T = 0$ on every representation (Theorem 3), and on $\mathbb{C}[G]$: $0 = T e_e = \sum_g f(g) e_g$, so $f = 0$. Thus the orthonormal family of Corollary 11(a) spans; class functions are functions on the set of classes, which gives the dimension count. $\square$

Lemma 14. If K is the conjugacy class of g then $|K| |C_G(g)| = |G|$. L-3

Proof. $hgh^{-1} = h'gh'^{-1}$ iff $h' \in hC_G(g)$, so $h \mapsto hgh^{-1}$ induces a bijection from the left cosets of $C_G(g)$ to K ; there are $|G|/|C_G(g)|$ cosets. $\square$

Theorem 15 (column orthogonality). $\sum_{i=1}^h \chi_i(g) \overline{\chi_i(g')} = |C_G(g)|$ if g, g' are conjugate, and 0 otherwise. T-5

Proof. Let $g_1, \dots, g_h$ represent the classes $K_1, \dots, K_h$ and $X_{ij} = \chi_i(g_j) \sqrt{|K_j|/|G|}$, a square matrix by Theorem 13. Row orthonormality reads $XX^* = I$, hence $X^*X = I$, i.e. $\sum_i \chi_i(g_j) \overline{\chi_i(g_l)} = \delta_{jl} |G|/|K_j| = \delta_{jl} |C_G(g_j)|$ by Lemma 14. $\square$

3.5 Certified character tables

Remark 1 (how an example is certified). For a group given by generators, the verifier computes the multiplication table, inverses and conjugacy classes, and for a claimed list of representations checks in exact arithmetic: (1) each map is a homomorphism on all $|G|^2$ pairs; (2) the Gram matrix $(\langle \chi_i, \chi_j \rangle)$ is the identity, so the representations are irreducible and pairwise non-isomorphic

(Corollary 11(d),(e)); (3) $\sum_i (\dim V_i)^2 = |G|$, so the list is complete (Proposition 12(d)). Steps (1)–(3) prove the table. As independent consistency checks it also tests h = number of classes (Theorem 13), column orthogonality with $|C_G(g)|$ (Theorem 15) and $\chi_{\text{reg}} = \sum_i \dim V_i \chi_i$ (Proposition 12). The tables below are typeset directly from the verifier’s output.

Proposition 16. *Let $C_n = \langle c \rangle$ and $\zeta = e^{2\pi i/n}$. The irreducible representations of C_n are $\chi_j(c^k) = \zeta^{jk}$, $j = 0, \dots, n-1$.* P-3

Proof. Each χ_j is a well-defined homomorphism since $\zeta^n = 1$; it is irreducible of dimension 1. The values $\chi_j(c) = \zeta^j$ are distinct, so the χ_j are pairwise non-isomorphic (Proposition 2(v)), and $n \cdot 1^2 = |C_n|$ shows completeness (Proposition 12(d)). □

Example 17. The tables of Proposition 16 are certified in $\mathbb{Q}(\zeta_n)$ for $n = 1, \dots, 8$. For $n = 4$:

class	e	c	c^2	c^3
size	1	1	1	1
$ C_G(g) $	4	4	4	4
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

E-C1, E-C2,
E-C3, E-C4,
E-C5, E-C6,
E-C7, E-C8

Example 18. S_3 : trivial, sign, and the standard representation on $\{x \in \mathbb{C}^3 : \sum x_i = 0\}$. By hand: $\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle = (4 + 0 + 2)/6 = 1$ and $1 + 1 + 4 = 6$. E-S3

class	e	(12)	(123)
size	1	3	2
$ C_G(g) $	6	2	3
trivial	1	1	1
sign	1	-1	1
standard	2	0	-1

Example 19. $D_4 = \langle r, s \rangle$, the symmetries of a square (r a rotation by $\pi/2$, s a reflection). The four one-dimensional representations $\varepsilon_{\pm\pm}$ send r, s to ± 1 ; “plane” sends $r \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $s \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. By hand: $\langle \chi, \chi \rangle = (4 + 4)/8 = 1$ for the plane, $4 + 4 = 8$. E-D4

class	e	r	s	r^2	rs
size	1	2	2	1	2
$ C_G(g) $	8	4	4	8	4
ε_{++}	1	1	1	1	1
ε_{+-}	1	1	-1	1	-1
ε_{-+}	1	-1	1	1	-1
ε_{--}	1	-1	-1	1	1
plane	2	0	0	-2	0

Example 20. $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ realized by $i \mapsto \text{diag}(i, -i)$, $j \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; $\varepsilon_{\pm\pm}$ send i, j to ± 1 . By hand: $\langle \chi, \chi \rangle = (4 + 4)/8 = 1$ for the quaternion representation, $4 + 4 = 8$. E-Q8

class	1	i	j	-1	k
size	1	2	2	1	2
$ C_G(g) $	8	4	4	8	4
ε_{++}	1	1	1	1	1
ε_{+-}	1	1	-1	1	-1
ε_{-+}	1	-1	1	1	-1
ε_{--}	1	-1	-1	1	1
quaternion	2	0	0	-2	0

Example 21. A_4 with $\omega = e^{2\pi i/3}$: λ_j factor through $A_4 \rightarrow C_3$, $(123) \mapsto \omega^j$, $(12)(34) \mapsto 1$; E-A4
the standard representation is the restriction of the one of S_4 . By hand: $(9 + 3)/12 = 1$ and $1 + 1 + 1 + 9 = 12$.

class	e	(123)	$(12)(34)$	(132)
size	1	4	3	4
$ C_G(g) $	12	3	4	3
λ_0	1	1	1	1
λ_1	1	ω	1	ω^2
λ_2	1	ω^2	1	ω
standard	3	0	-1	0

Example 22. S_4 : σ_2 is the standard representation of S_3 composed with $S_4 \rightarrow S_3$, the action on E-S4
the three pairings $\{12|34, 13|24, 14|23\}$. By hand: $(9 + 6 + 6 + 0 + 3)/24 = 1$ for the standard representation and $1 + 1 + 4 + 9 + 9 = 24$.

class	e	(12)	(1234)	(123)	$(12)(34)$
size	1	6	6	8	3
$ C_G(g) $	24	4	4	3	8
trivial	1	1	1	1	1
sign	1	-1	-1	1	1
σ_2	2	0	0	-1	2
standard	3	1	-1	0	-1
standard $\otimes$ sign	3	-1	1	0	-1

3.6 The character table does not determine the group

Proposition 23. D_4 and Q_8 have the same character table, but are not isomorphic. E-DQ

Proof. The tables in Examples 19 and 20 agree under $e \leftrightarrow 1$, $r \leftrightarrow i$, $s \leftrightarrow j$, $r^2 \leftrightarrow -1$, $rs \leftrightarrow k$, which respects class sizes. An isomorphism preserves element orders, but D_4 has five elements of order 2 (r^2 and four reflections) and Q_8 only one (-1). *Certificate.* The verifier searches all size-preserving class bijections for one under which the tables agree as sets of rows, and counts elements of order 2 from the multiplication tables (5 and 1). In its self-test the same procedure correctly reports that C_4 and $C_2 \times C_2$ have different tables. $\square$

Remark 2 (trusted base). The computer-assisted parts rely on Python's `fractions.Fraction` (exact rationals) and a small implementation of $\mathbb{Q}(\zeta_N) = \mathbb{Q}[x]/(\Phi_N)$ in which equality is decided by the remainder modulo the cyclotomic polynomial Φ_N (computed exactly from $x^N - 1 =$

$\prod_{d|N} \Phi_d$). No floating-point number is used. A third party re-runs everything with `python3 repththeory/certify.py`. The general proofs of this paper are checked by reading, not by a proof assistant.

4 Scope and omitted topics

To keep the text self-contained we omit: (i) that $\dim V_i$ divides $|G|$, which needs algebraic integers; (ii) induced representations and Frobenius reciprocity; (iii) the Frobenius–Schur indicator and real representations; (iv) positive characteristic, where Theorem 3 fails when the characteristic divides $|G|$. A natural next step is to formalize Lemma 7 and Theorem 9 in a proof assistant, which would lift those claims from `proved-text` to a machine-checked level.

Declarations

AI use. Text, proofs and verifier were drafted with an AI assistant inside a verifier-gated workflow; every claim is either recomputed exactly by the verifier or proved in the text, and the build refuses to typeset a theorem-like environment without a backed claim. **Code and data.** `repththeory/` in the project repository: `certify.py`, `cyclo.py`, `build_paper.py`, the plain-text extraction `theorems.txt` and the tables `out/claims.csv`, `out/gates.csv`. **Competing interests.** None.

References

- [1] W. Fulton and J. Harris, *Representation Theory*, Springer New York, 2004. doi:10.1007/978-1-4612-0979-9.
- [2] G. James and M. Liebeck, *Representations and Characters of Groups*, Cambridge University Press, 2001. doi:10.1017/CBO9780511814532.
- [3] J.-P. Serre, *Linear Representations of Finite Groups*, Springer New York, 1977. doi:10.1007/978-1-4684-9458-7.

A Verifier self-test

Before any certificate is computed, the verifier judges 15 statements with known truth value; a single wrong verdict aborts the run. True: the list $\{\text{trivial, sign, standard}\}$ of S_3 is complete; $c \mapsto i$ defines a representation of C_4 ; $\dim V^G = 1$ for S_4 on four points; D_4 and Q_8 share a table and are not isomorphic; $1 + \omega + \omega^2 = 0$; $i\bar{i} = 1$; $1 + i + i^2 + i^3 = 0$ (a regression case for a canonical-form error found while building the verifier). False: the S_3 list with the reducible permutation representation; the S_4 list without σ_2 ($\sum d^2 = 20$); $r \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $s \mapsto \text{diag}(1, -1)$ for D_4 ; $c \mapsto \zeta_8$ for C_4 ; “ S_4 on four points has two orbits”; C_4 and $C_2 \times C_2$ share a table; $1 + \omega = 0$; $i^2 = 1$. All 15 verdicts are correct (gate `G0_selftest`).

B Provenance

Every claim, its level and its evidence, as written by the verifier to `out/claims.csv`. Level `proved-text`: proof in this paper and in `theorems.txt`; the evidence column lists the certificates

that test the statement on examples. Level `computed-rigorous`: exact recomputation by the named function of `certify.py`.

claim	level	status	evidence
E-C1	computed-rigorous	verified	certify.cert_character_table
E-C2	computed-rigorous	verified	certify.cert_character_table
E-C3	computed-rigorous	verified	certify.cert_character_table
E-C4	computed-rigorous	verified	certify.cert_character_table
E-C5	computed-rigorous	verified	certify.cert_character_table
E-C6	computed-rigorous	verified	certify.cert_character_table
E-C7	computed-rigorous	verified	certify.cert_character_table
E-C8	computed-rigorous	verified	certify.cert_character_table
E-S3	computed-rigorous	verified	certify.cert_character_table
E-D4	computed-rigorous	verified	certify.cert_character_table
E-Q8	computed-rigorous	verified	certify.cert_character_table
E-A4	computed-rigorous	verified	certify.cert_character_table
E-S4	computed-rigorous	verified	certify.cert_character_table
E-T3-S3	computed-rigorous	verified	certify.cert_orthogonality_vs_linear_algebra
E-T3-S4	computed-rigorous	verified	certify.cert_orthogonality_vs_linear_algebra
E-L2-S3	computed-rigorous	verified	certify.cert_fixed_points
E-L2-D4	computed-rigorous	verified	certify.cert_fixed_points
E-L2-A4	computed-rigorous	verified	certify.cert_fixed_points
E-L2-S4	computed-rigorous	verified	certify.cert_fixed_points
E-L2-S3x	computed-rigorous	verified	certify.cert_fixed_points
E-M	computed-rigorous	verified	certify.cert_maschke_example
E-DQ	computed-rigorous	verified	certify.cert_same_table_not_isomorphic
L-1	proved-text	proved	theorems.txt
P-1	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8 E-S3 E-D4 E-Q8 E-A4 E-S4
T-1	proved-text	proved	theorems.txt + E-M
T-2	proved-text	proved	theorems.txt + E-T3-S3 E-T3-S4
C-1	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8
L-2	proved-text	proved	theorems.txt + E-L2-S3 E-L2-D4 E-L2-A4 E-L2-S4 E-L2-S3x
T-3	proved-text	proved	theorems.txt + E-T3-S3 E-T3-S4
C-2	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8 E-S3 E-D4 E-Q8 E-A4 E-S4
P-2	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8 E-S3 E-D4 E-Q8 E-A4 E-S4
T-4	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8 E-S3 E-D4 E-Q8 E-A4 E-S4
L-3	proved-text	proved	theorems.txt
T-5	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8 E-S3 E-D4 E-Q8 E-A4 E-S4
P-3	proved-text	proved	theorems.txt + E-C1 E-C2 E-C3 E-C4 E-C5 E-C6 E-C7 E-C8